

Comprehensive Report on Predictive Modeling for Tumor Marker Scores

Introduction

This project aims to develop an accurate predictive model to understand and predict tumor marker scores (CEA and CA2729) based on participant's vocal and speech patterns recorded during storyline sessions. The objective is to leverage language models to generate additional relevant features from the participants' text answers to enhance the predictive power of the models.

Data Description

df_storyline_data.csv: Contains assessment IDs, dates of storyline sessions, and data on facial, speech, and vocal patterns exhibited by participants.

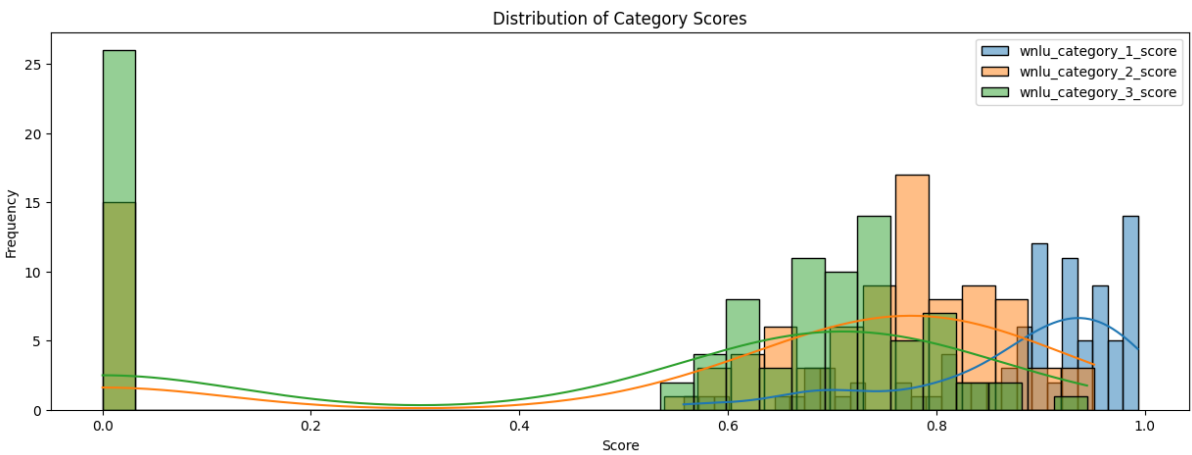
df_tumormarker_data.csv: Contains clinical tumor marker test results, including the date of the test, and scores for CEA and CA2729 tumor markers

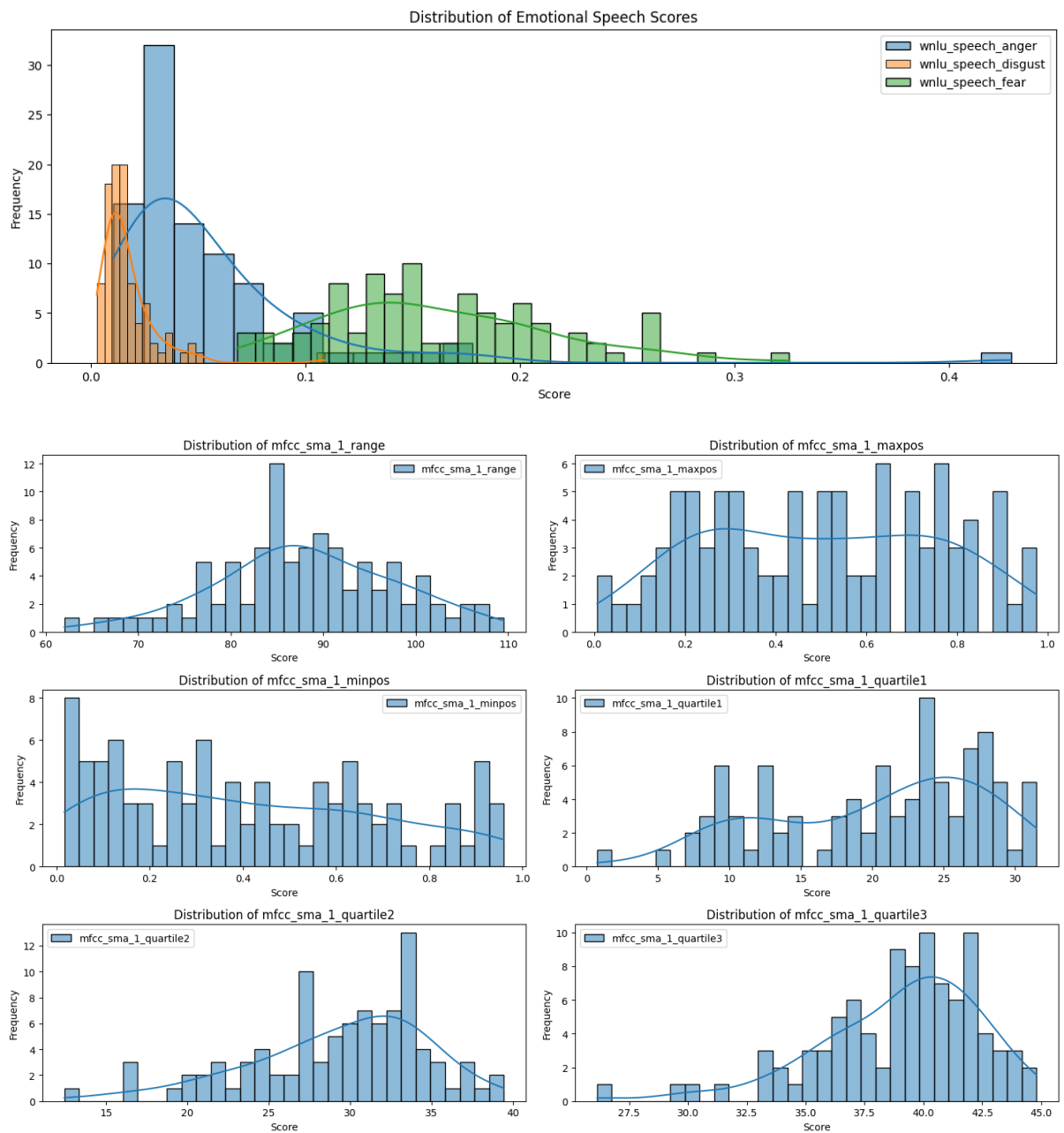
Key Findings from Data Exploration

Frequency of Data Collection: Storyline sessions are conducted every two days, whereas tumor marker scores are recorded once a week.

Initial Data Patterns: Initial exploration indicated a need to align the more frequent storyline data with the weekly tumor marker test dates.

The exploratory data analysis provided crucial insights into the distribution and variability of key features within the dataset. These insights guide the feature selection and engineering process, ensuring that the most relevant and informative features are used in the predictive models. Understanding the distribution of category scores, emotional speech scores, and MFCC features helps in identifying relevant patterns and characteristics that could improve the accuracy and robustness of the models.





Feature Engineering Process

Generating Features Using Language Models

Tools Utilized: GPT-3 and BERT language models were employed to extract meaningful features from the text responses.

Procedure: Sentiment Analysis: Extracted sentiment scores from the speech text.

Topic Modeling: Identified and scored topics and categories discussed by participants.

Emotion Detection: Detected emotions such as anger, joy, and fear in the speech patterns.

Example Features

wnlu_category_1_score: A score indicating the presence of specific categories derived from the language model analysis.

mfcc_sma_13_percentile1_0: Represents a percentile of the 13th Mel-Frequency Cepstral Coefficient, capturing timbral aspects of the audio.

Modeling Approach and Evaluation Results

Data Preparation

Aggregation: Storyline data was aggregated to a weekly summary to match the frequency of tumor marker tests.

Feature Selection: Important features were selected based on their relevance to the target variables (CEA and CA2729).

Models Evaluated

This project compared baseline models, which used the original dataset, with enhanced models incorporating additional features generated by advanced language models like GPT-3 and BERT. The performance of these models was evaluated using three primary metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R²).

Baseline Models: Trained on the original dataset without additional features.

Enhanced Models: These features are generated using language models.

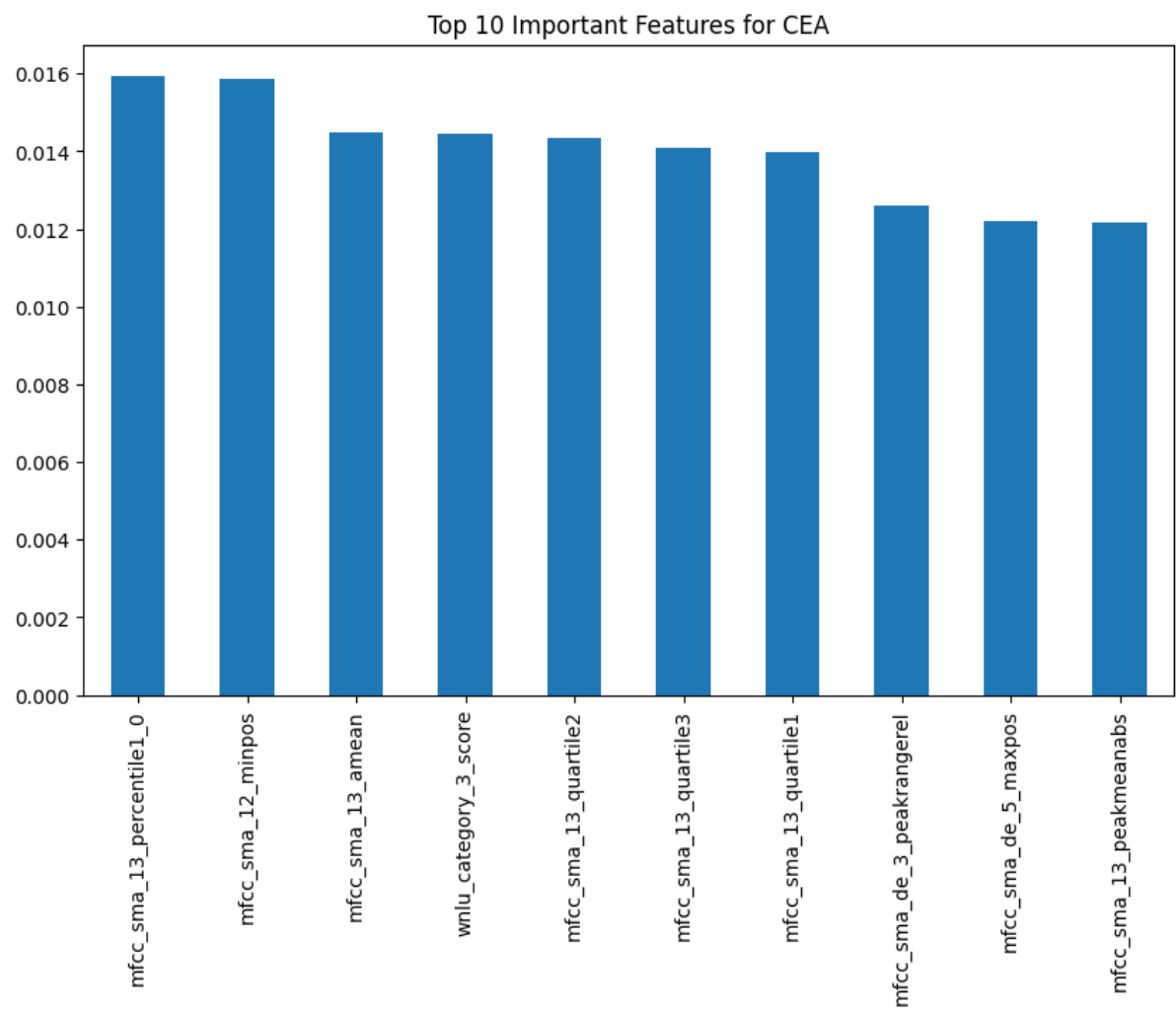
Results

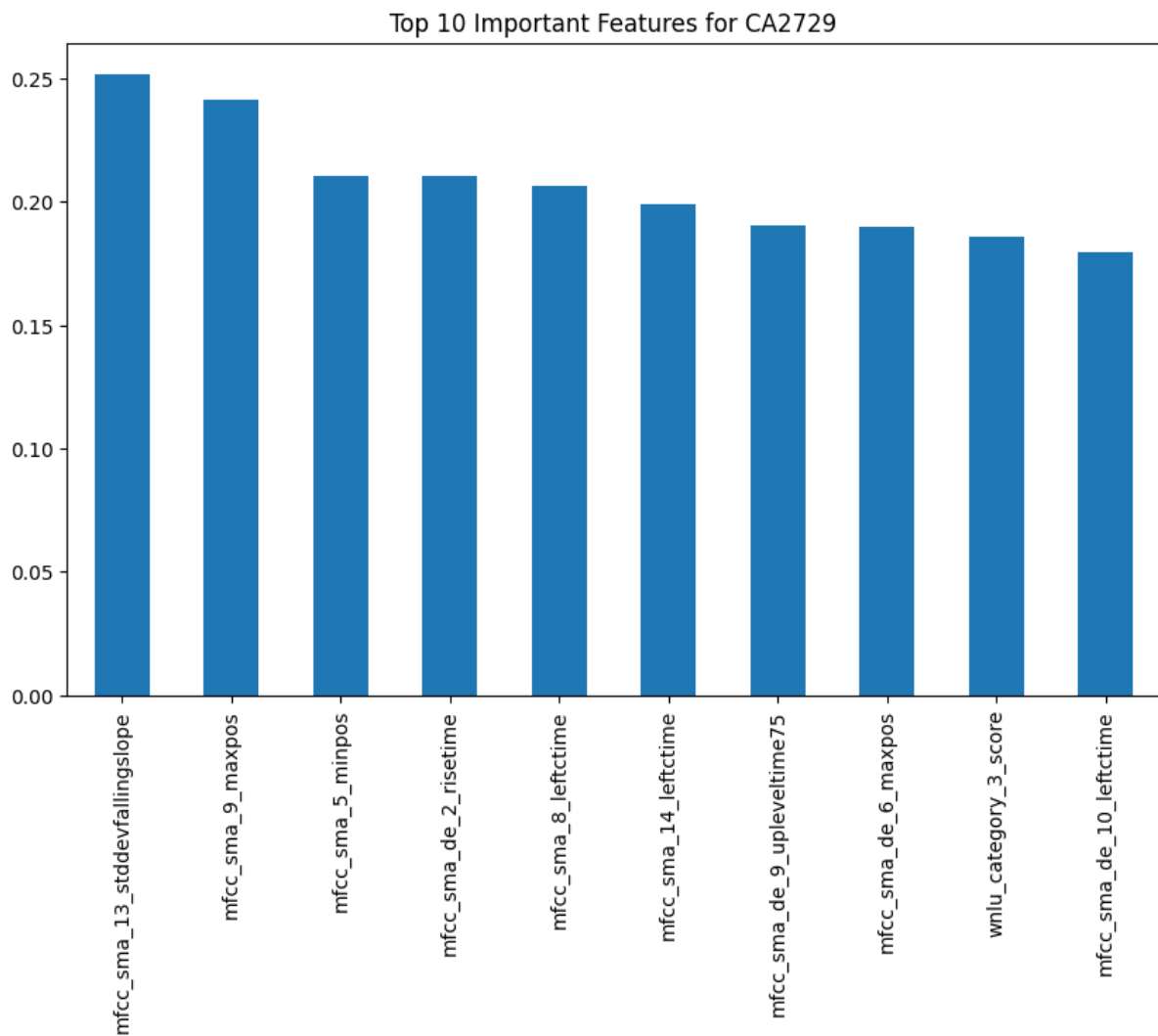
For this project, I chose Gradient Boosting as the main modeling technique because of its powerful ability to make accurate predictions and effectively manage complex interactions between different features. The vocal and speech patterns in our data are quite intricate and varied, and since our goal was to predict tumor marker scores (CEA and CA2729) with high accuracy, Gradient Boosting was a natural fit. It’s particularly well-suited for capturing the subtle relationships within the dataset, making it an ideal choice for this challenging task.

Model	MAE	MSE	R²
Original Dataset - CEA	0.36	0.18	0.33
Original Dataset - CA2729	6.05	67.77	-0.23
LLM - CEA	0.49	0.36	-0.20
LLM - CA2729	5.83	48.88	-0.44

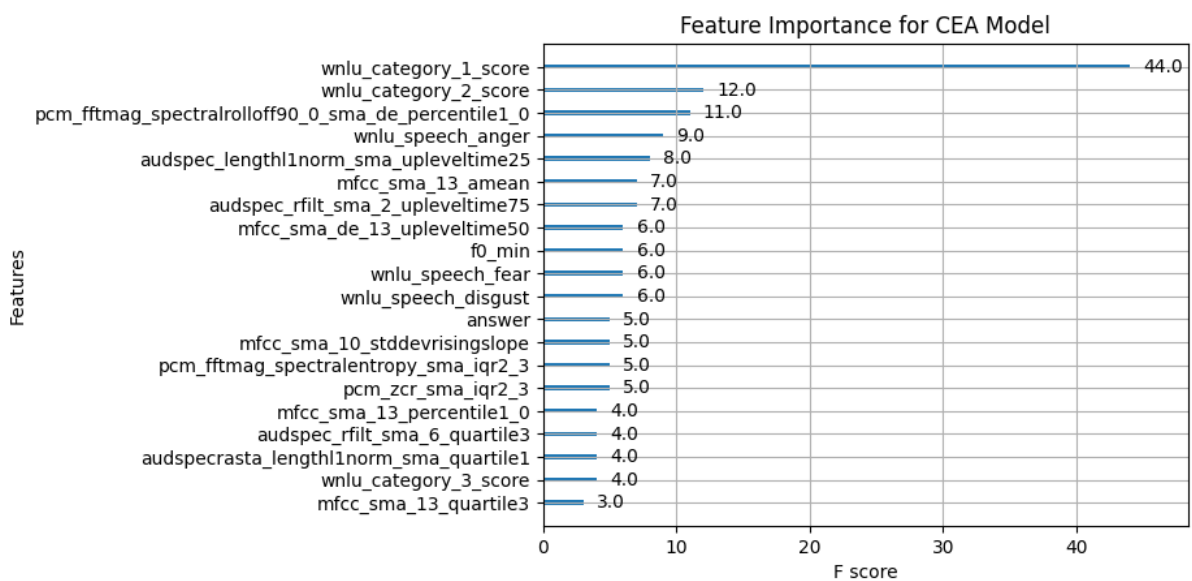
BERT - CEA	0.53	0.37	-0.33
BERT - CA2729	5.74	44.91	0.19

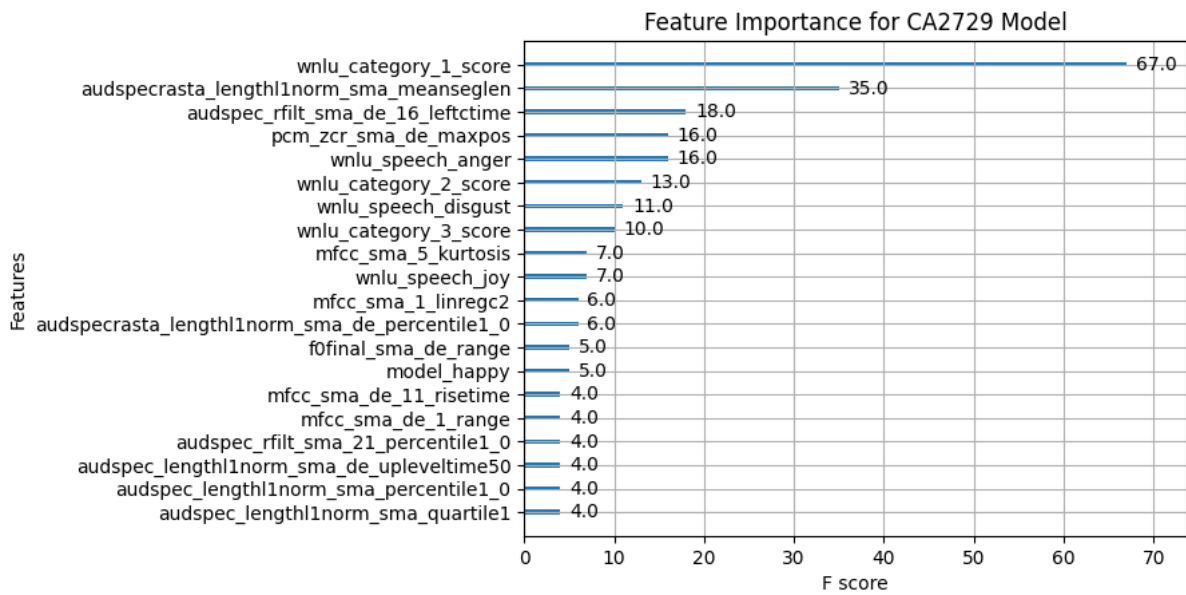
Visualizations: Important Features



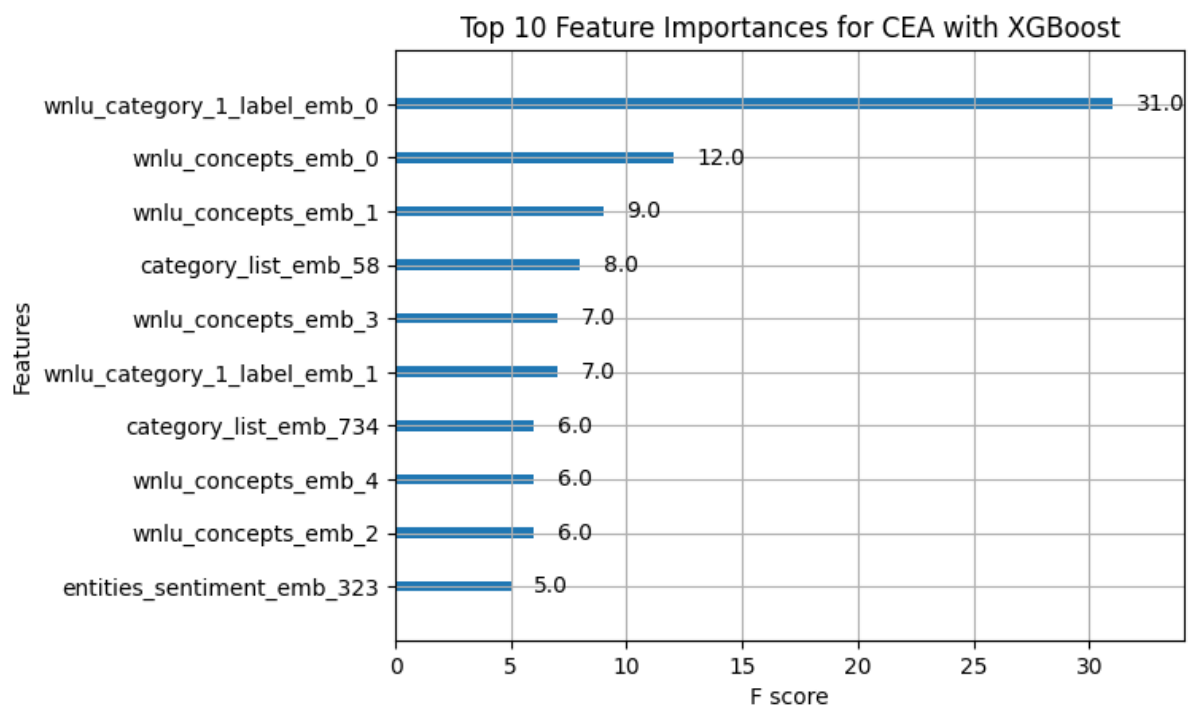


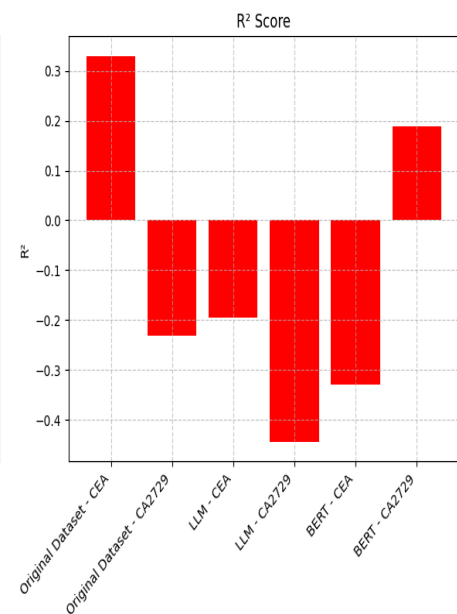
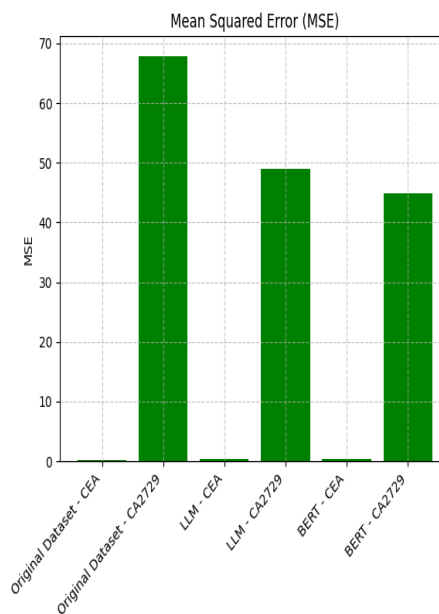
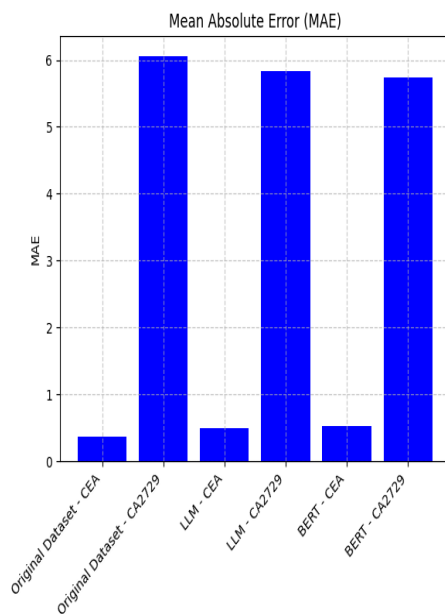
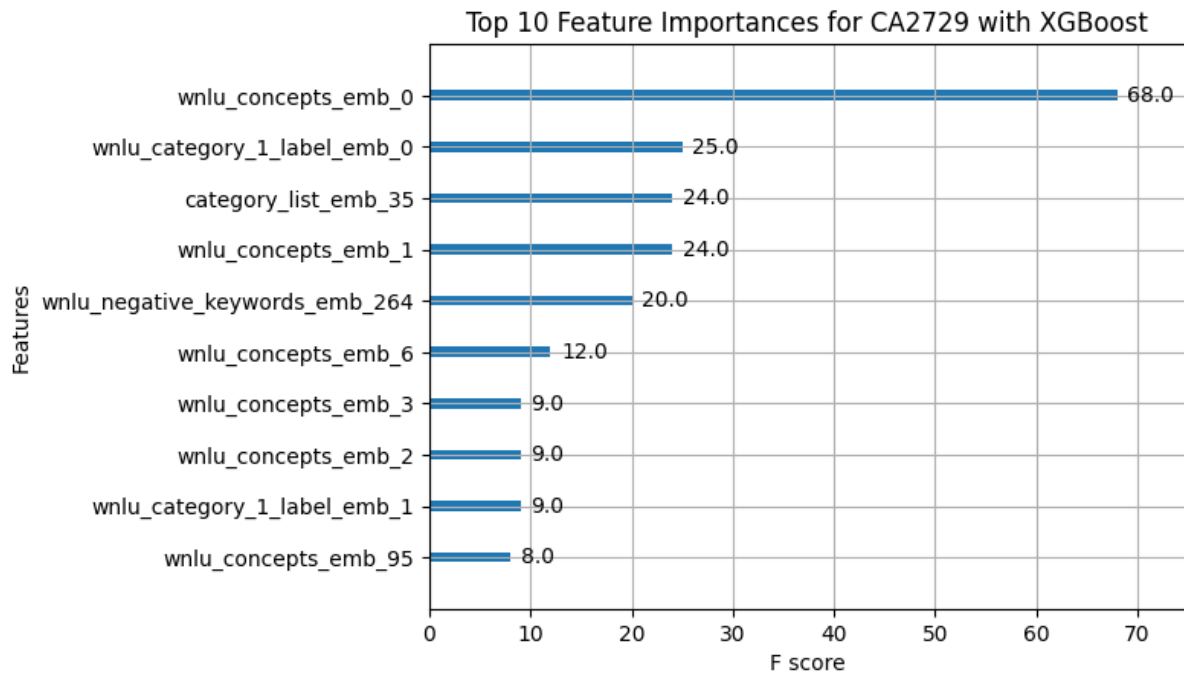
Important Features for OpenAI gpt-3(langchain) :





Important Features for BERT :





Importance of Features:

The analysis revealed that advanced feature engineering using language models like GPT-3 and BERT can significantly impact model performance. While not all generated features were beneficial, the right set of features, particularly those capturing semantic and emotional nuances from speech, showed potential for improving predictive accuracy.

Enhanced models demonstrated mixed results. For example, BERT-enhanced models showed improved performance for predicting CA2729 tumor marker scores, with a notable positive R^2 value, indicating a better fit compared to baseline models. Conversely, the additional features did not enhance CEA predictions, sometimes even increasing prediction errors.

The varying impact of language model features on different tumor markers suggests that feature relevance is highly context-specific. Features that improved CA2729 predictions did not necessarily help with CEA, highlighting the need for tailored feature engineering and model development for different predictive targets.

Key Limitations Observed

Challenge: Storyline sessions were conducted every two days, while tumor marker scores were recorded weekly. Aligning these datasets required aggregating the more frequent storyline data to match the weekly tumor marker tests.

Impact: This aggregation potentially led to the loss of detailed temporal information, which could affect the model's ability to capture nuanced patterns.

Challenge: Not all features generated by the language models contributed positively to the prediction accuracy. For example, additional features did not improve the model performance for CEA prediction and, in some cases, led to increased errors.

Impact: This highlighted the need for careful feature selection and validation to ensure that only relevant and effective features are included in the model.

Challenge: The models' performance might vary with different datasets, indicating potential issues with generalization. The mixed results between CEA and CA2729 predictions suggest that the models might not be robust across different contexts.

Impact: Ensuring that models can generalize well to new and unseen data is crucial for practical applications, necessitating further validation and possibly more diverse training data.

Challenge: The relatively infrequent collection of tumor marker scores (once a week) may not capture rapid changes in the patient's condition, limiting the model's responsiveness to such variations.

Impact: More frequent data collection could provide a richer dataset, improving the model's ability to detect and predict subtle changes over time.

Challenge: Some enhanced models, especially those using language model features, showed negative R^2 values, indicating poor fit and predictive power.

Impact: Negative R^2 values for models predicting CEA suggested that the additional features might introduce noise rather than useful information, emphasizing the importance of model tuning and feature engineering

Next Steps and Improvements

Enhanced Data Collection: Collecting more frequent and detailed storyline session data.

Advanced Feature Engineering: Exploring other advanced language models and feature extraction techniques.

Model Tuning and Ensemble Methods: Further hyperparameter tuning and exploring ensemble methods to improve predictive performance.